Qatar-1b

R-1531

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_Qatar-1_190416` · created 2026-08-01T18:06:25+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458589.9211 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.057
Transit depth (Rp/Rs)^2	3.67 +/- 2.17 [%]
Orbital Inclination (inc)	87.99 +/- 2.9
Airmass coefficient 1 (a1)	0.081 +/- 0.046
Airmass coefficient 2 (a2)	1.72 +/- 0.67
Scatter in the residuals of the lightcurve fit is	62.02 %
Best Comparison Star	#3 - [531, 67]
Optimal Aperture	2.05
Optimal Annulus	5.5
Transit Duration (day)	0.0818 +/- 0.009

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.192 ± 0.057 vs 0.1463 (deviation 0.68σ)
Duration fit vs published	0.0818 d vs 0.0692 d (deviation 1.19σ)
Mid-time O–C	-3.7 min
Depth SNR	2.9
Residual scatter	62.02 %

Light curve

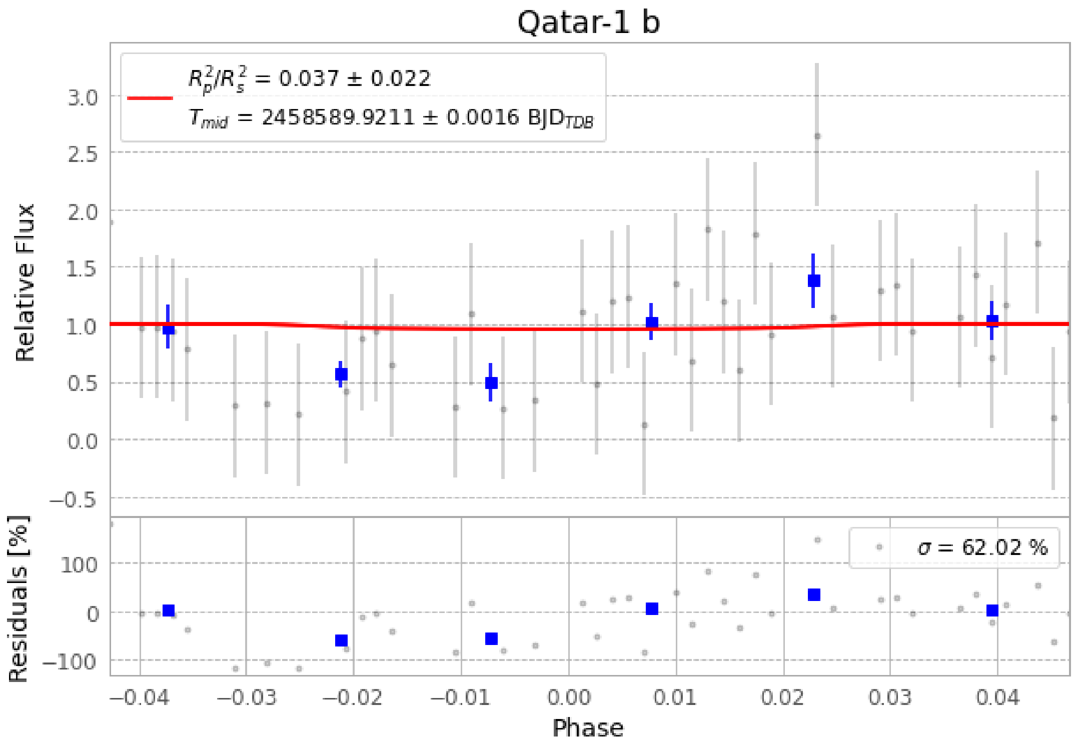

Figure 1 — FinalLightCurve_Qatar-1 b_2019-04-16.png (SHA-256 e5dedbc4ce46a204…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [306, 351], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 553fc4cee422366f…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

63 FITS frames (41 MB), `Qatar-1190416083912.FITS` → `Qatar-1190416114512.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-190416.log (dfcdc5944007…), FinalLightCurve_Qatar-1 b_2019-04-16.png (e5dedbc4ce46…), FinalLightCurve_Qatar-1 b_2019-04-16.csv (8ea02fcaab94…)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-08-01 18:06Z.